

Advanced Power BI Interview Questions with Detailed Answers

These are the types of questions commonly asked for Data Analyst, Business Intelligence Developer, and Power BI Developer roles.

1. What is the difference between a Measure, Calculated Column, and Calculated Table?

Answer

A **Calculated Column** is computed row by row during data refresh and stored in the model, which increases model size.

A **Measure** is calculated dynamically at query time based on the current filter context, making it memory efficient.

A **Calculated Table** creates an entirely new table using DAX.

Example

```
Profit Margin =  
DIVIDE(  
    SUM(Sales[Profit]),  
    SUM(Sales[SalesAmount]),  
    0  
)
```

Best Practice: Use measures whenever possible.

2. Explain Row Context and Filter Context.

Answer

- **Row Context** exists when Power BI evaluates each row individually, such as in calculated columns or iterator functions like SUMX.
- **Filter Context** is created by report filters, slicers, rows, columns, and CALCULATE.

Understanding these two contexts is essential for mastering DAX.

3. What does CALCULATE() do?

Answer

CALCULATE modifies the existing filter context before evaluating an expression.

Sales for USA =

```
CALCULATE(  
    SUM(Sales[SalesAmount]),  
    Customer[Country] = "USA"  
)
```

CALCULATE is considered the most powerful DAX function.

4. What is Context Transition?

Answer

Context Transition occurs when CALCULATE converts row context into filter context.

This typically happens inside calculated columns or iterator functions.

5. Explain Query Folding.

Answer

Query Folding pushes transformation logic back to the source database instead of processing it in Power BI.

Benefits include:

- Faster refresh
 - Reduced memory usage
 - Better scalability
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6. How do you identify whether Query Folding is working?

Answer

In Power Query:

- Right-click a step
- Select **View Native Query**

If available, query folding is active.

7. What is the difference between Import Mode, DirectQuery, and Live Connection?

Answer

Mode	Storage	Performance	Use Case
Import	In-memory	Fastest	Most scenarios
DirectQuery	Source database	Moderate	Real-time data
Live Connection	External semantic model	Depends	Enterprise models

8. What is Composite Model?

Answer

Composite Model allows combining Import and DirectQuery tables in a single Power BI model.

Useful when balancing performance and real-time requirements.

9. What are Aggregations in Power BI?

Answer

Aggregations are pre-summarized tables used to improve query performance over large datasets.

They reduce DirectQuery hits significantly.

10. What is Incremental Refresh?

Answer

Incremental Refresh refreshes only new or changed data instead of reloading the entire dataset.

Ideal for large fact tables.

11. What is the VertiPaq Engine?

Answer

VertiPaq is the in-memory columnar storage engine used in Import mode.

It provides high compression and extremely fast query performance.

12. How does Cardinality affect performance?

Answer

Higher cardinality reduces compression efficiency and can slow down performance.

Columns with unique values should be minimized where possible.

13. Explain Star Schema and why it is preferred.

Answer

A Star Schema consists of:

- One fact table
- Multiple dimension tables

Benefits:

- Better performance
 - Easier DAX
 - Improved maintainability
-

14. Difference between USERELATIONSHIP and CROSSFILTER.

Answer

- **USERELATIONSHIP** activates an inactive relationship temporarily.
- **CROSSFILTER** changes filter direction during calculation.

Sales by Ship Date =

```
CALCULATE(  
    SUM(Sales[Amount]),  
    USERELATIONSHIP(Sales[ShipDate], Date[Date])  
)
```

15. What is Bidirectional Filtering, and when should it be avoided?

Answer

Bidirectional filtering propagates filters in both directions.

Avoid overusing it because it can:

- Create ambiguity
 - Slow performance
 - Produce unexpected results
-

16. Explain Dynamic Row-Level Security.

Answer

Dynamic RLS filters data based on the logged-in user.

[Email] = USERPRINCIPALNAME()

17. What is the difference between ALL(), ALLEXCEPT(), and ALLSELECTED()?

Answer

- **ALL()** removes all filters.
 - **ALLEXCEPT()** removes all except specified columns.
 - **ALLSELECTED()** respects external filters while ignoring internal visual filters.
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18. How would you optimize a slow Power BI report?

Answer

- Use Star Schema
 - Remove unused columns
 - Reduce visual count
 - Optimize DAX
 - Avoid bidirectional relationships
 - Use aggregations
 - Enable incremental refresh
-

19. What is Auto Date/Time, and why should it be disabled?

Answer

Auto Date/Time automatically creates hidden date tables.

It increases model size and should be disabled in production.

20. How do you troubleshoot refresh failures?

Answer

- Review refresh history
 - Check gateway status
 - Verify credentials
 - Inspect source changes
 - Test Power Query steps
-

21. What are Calculation Groups?

Answer

Calculation Groups reduce repetitive measure creation, especially for time intelligence.

They are created using external tools like Tabular Editor.

22. What is the difference between SUMMARIZE and SUMMARIZECOLUMNS?

Answer

SUMMARIZECOLUMNS is optimized for queries and generally preferred over SUMMARIZE.

23. What is the role of Tabular Editor?

Answer

Tabular Editor is used for:

- Creating calculation groups
- Managing measures
- Best Practice Analyzer

- Automating model changes
-

24. What is a Dataflow?

Answer

A Dataflow is a cloud-based ETL process in Microsoft Power BI Service that promotes reusable transformations.

25. Explain Deployment Pipelines.

Answer

Deployment Pipelines help move content across:

- Development
- Test
- Production

They support controlled release management.

26. What is XMLA Endpoint?

Answer

XMLA Endpoint enables external tools to connect to Power BI semantic models.

Used for enterprise-grade administration and automation.

27. What is a Semantic Model?

Answer

A semantic model is the modern name for a Power BI dataset.

It contains:

- Tables
- Relationships
- Measures
- Security rules

28. Difference between Dashboard and Report.**Answer**

- Dashboard: Single-page, created in Service.
- Report: Multi-page, created in Desktop.

29. How do you handle Many-to-Many relationships?**Answer**

Use a bridge table to resolve ambiguity and improve model performance.

30. What is the best way to model slowly changing dimensions?**Answer**

Typically:

- Type 2 dimension tables
- Surrogate keys
- Effective date ranges